

Summary

Master of Science in Machine Learning student at the University of Tübingen supported by the Research-Oriented Master's Scholarship of the Zuse School ELIZA, with a background in empirical machine-learning research and research engineering. My main research interests are deep learning theory and causality, particularly how optimization shapes learning dynamics and the representations neural networks learn, and when those representations capture the causal structure behind the data. I am moving toward theoretical work while bringing empirical intuition, controlled experimentation, and software engineering experience.

Education

2026– **Master of Science in Machine Learning**, *University of Tübingen*, Tübingen, Germany
Research-oriented, four-semester programme; **Master's Scholar, Zuse School ELIZA** (Research-Oriented Master's Scholarship). Focus on probability, statistics, optimization, and learning theory.

2025 **Exchange, Computer Science & Engineering**, *Hong Kong University of Science and Technology*, Hong Kong SAR
Courses in machine learning, deep learning in computer vision, large language models, reinforcement learning, and postgraduate data visualisation.

2023–2026 **Bachelor of Science in Engineering, Software Engineering**, *University of Southern Denmark*, Sønderborg, Denmark
Graduated June 2026. Danish weighted average: **10.71/12** over 140 numerically graded ECTS (approximately **3.71/4.0**, WES-style U.S. equivalent). Co-authored the bachelor thesis with Tim Lukas Adam (equal contribution); **selected as the programme's best in a graduating cohort of more than 100 students**.

Research Experience

Oct 2026– **Research Collaborator**, *Applied and Interpretable Machine Learning Research Group, University of Southern Denmark*, Remote
Continue to conduct research with the group after completing the BSc, as a member of its research team with access to its compute resources.

Jan–Oct 2026 **Undergraduate Student Researcher**, *Applied and Interpretable Machine Learning Research Group, University of Southern Denmark* (Supervisor: Serkan Ayvaz), Sønderborg, Denmark
Independently initiated and led research projects, owning problem formulation, experimental design, implementation, analysis, and manuscript preparation; first author on multiple resulting publications.

Dec 2025–Sep 2026 **Research Collaborator**, *Hong Kong University of Science and Technology DataVISards* (Supervisor: Arpit Narechania), Hong Kong SAR
Contributed to a collaborative visualisation research project established during a 2025 exchange semester, resulting in a paper accepted at IEEE VIS 2026.

Sep 2024–Dec 2025 **Student Research Assistant**, *University of Southern Denmark Data & Intelligence Lab*, Sønderborg, Denmark
Built machine-learning pipelines and contributed to study design, literature review, manuscript preparation, grant proposals, and multimodal dataset management across lab projects, alongside my own first-author research.

Selected Publications

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Routing Subspaces: Auditing Evaluation-to-Deployment Mismatch in Fine-Tuned Language Models*. *Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted. [arXiv]

2026 **P. M. Konrad**, A.-A. Popa, Y. Sabzehmeidani, L. Zhong, M. Tripathy, A. Constantinescu, E. A. Liehn, S. Ayvaz. *Challenges in Deep Learning-Based Small Organ Segmentation: A Benchmarking Perspective for Medical Research with Limited Datasets*. *Biomedical Signal Processing and Control* 128, 111226 (Q1, SJR). [Published version] [arXiv]
Ten-model benchmark with ablations, bootstrap uncertainty, and independent distribution-shift evaluation for small cardiovascular histology datasets.

2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Self-Reports Do Not Identify Self-Models: An Identifiability Test for Counterfactual Reports*. Oral presentation, *Philosophy of Machine Learning Workshop at the International Conference on Machine Learning*. [OpenReview]
Environment-shift invariance tests across three open models separate report accuracy from mechanism binding.

Selected Publications (continued)

- 2026 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Decoded but Unused: Instruction Tuning Routes Moral Framing into the Judgment Readout*. Poster, *Workshop on Mechanistic Interpretability at the International Conference on Machine Learning*. [OpenReview]
Matched pretrained/instruction-tuned models with probes, activation patching, head ablations, subspace controls, and cross-model replication.
- 2025 **P. M. Konrad**, T. Tanyel, S. Ayvaz. *Beyond Major Floods: Deep Learning for Detecting Shallow Water Inundation in Agricultural Areas*. *Knowledge-Based and Intelligent Information & Engineering Systems 2025; Procedia Computer Science 270*, 301–310. [Published version]

Teaching

- Jan–Aug 2026 **Teaching Assistant, Artificial Intelligence**, *University of Southern Denmark*
Led exercise sessions, supported students with Artificial Intelligence and Machine Learning concepts, helped design hands-on lab assignments, and developed interactive supplementary course material.

Invited Talks

- Feb 2026 **Agentic Engineering in Practice**, House of Software Sønderborg inaugural event; invited talk to 100+ attendees.

Academic Service

- 2025–2026 **Student Member**, Educational Committee, Software Engineering Programme, University of Southern Denmark. Represented students in curriculum and quality-assurance discussions concerning course content, assessment, and the study environment.

Selected Recognition

- 2026 **Best bachelor thesis**, Software Engineering programme, University of Southern Denmark; co-authored with Tim Lukas Adam (equal contribution).
- 2026 **Best Startup Award finalist** (one of five), DreamBear, SDU Startup Night.
- 2025 **Top 10**, Danish National Championship in Artificial Intelligence; individual competitor, organised by the Danish Society for Artificial Intelligence.
- 2024 **Top 10**, Danish National Championship in Artificial Intelligence; team competitor, organised by the Danish Society for Artificial Intelligence.

Military Service

- 2017–2021 **Staff Duty Soldier**, *Bundeswehr, German Armed Forces*, Glücksburg, Germany
Led a small HR team administering more than 600 soldiers. Received a **Formal Recognition for Exemplary Service** and a **Performance Bonus for Outstanding Achievement** in 2021.

Additional Experience

- Jan–Jun 2026 **Founder**, *SaturoLabs*, Denmark
Built and shipped DreamBear, an iOS app creating personalised narrated bedtime stories that frame neurodivergent traits as strengths.
- Jun–Nov 2024 **CTO & Co-Founder**, *Tutora ApS*, Sønderborg, Denmark
Led end-to-end development of the company's web application.
- 2022–2024 **Co-CEO & Co-Founder**, *Yeager GmbH*, Remote
Built an AI-assisted powerlifting training application.

Technical Skills

- Research Experimental design, controlled interventions, activation patching, representation probing, ablations, bootstrap uncertainty, model training and evaluation, segmentation, conformal prediction, reproducibility.
- Machine Learning Python, PyTorch, Hugging Face Transformers, TransformerLens, scikit-learn, pandas, NumPy, Optuna, Weights & Biases.
- Engineering Git, Docker, Linux, SQL, Google Cloud Platform, JavaScript.

Languages

- Fluent English, German
Basic Danish, Thai

Last updated 1 October 2026.